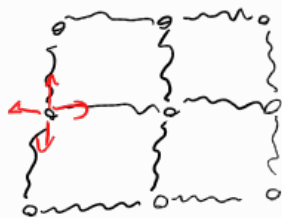


4.2) Einsteinov model trdnine, T visoka, $C_v = ?$

↙ krogični nihaj neodv. od ostalih



$$C_v = \left(\frac{\partial \langle E \rangle}{\partial T} \right)_v$$

$$\left\{ \begin{array}{l} 1D \text{ Harm. osc.} \\ E(n) = \hbar \omega (n + \frac{1}{2}) \\ n = 0, 1, \dots \end{array} \right.$$

C_u

$$\omega_e = 4 \cdot 10^{13} \text{ s}^{-1}$$

$$M = 64 \text{ kg/kmol}$$

$$T = 500 \text{ K}$$

$$C_v(+) = ?$$

en 1d HO:

$$\begin{aligned} e^{-\beta F} &= \sum_{n=0}^{\infty} e^{-\beta E_n} = \sum_{n=0}^{\infty} e^{-\beta \hbar \omega (n + \frac{1}{2})} = e^{-\frac{\beta \hbar \omega}{2}} \sum_{n=0}^{\infty} e^{-\beta \hbar \omega n} = \\ &= e^{-\frac{\beta \hbar \omega}{2}} \frac{1}{1 - e^{-\beta \hbar \omega}} = \frac{1}{e^{\frac{\beta \hbar \omega}{2}} - e^{-\frac{\beta \hbar \omega}{2}}} = \frac{1}{2 \text{sh}(\frac{\beta \hbar \omega}{2})} \end{aligned}$$

N 3D HO:

$$e^{-\beta F} = \left(2 \text{sh}(\frac{\beta \hbar \omega}{2}) \right)^{-3N}$$

$$\beta F = 3N \ln \left(2 \text{sh}(\frac{\beta \hbar \omega}{2}) \right)$$

$$\langle E \rangle = \frac{\partial (\beta F)}{\partial \beta} = 3N \frac{2 \text{ch}(\frac{\beta \hbar \omega}{2})}{2 \text{sh}(\frac{\beta \hbar \omega}{2})} \cdot \frac{\hbar \omega}{2} = \underline{3N \frac{\hbar \omega}{2} \text{cth}(\frac{\beta \hbar \omega}{2})}$$

$$\frac{1}{2} \langle \text{th} x \rangle = \frac{1}{2} \frac{e^x + e^{-x}}{e^x - e^{-x}} = \frac{1}{2} \frac{e^x - e^{-x} + 2e^{-x}}{e^x - e^{-x}} = \frac{1}{2} \left(1 + 2 \frac{e^{-x}}{e^x - e^{-x}} \right) = \frac{1}{2} + \frac{1}{e^{2x} - 1}$$

$$C_v = \frac{\partial \langle E \rangle}{\partial T} = 3N \left(\frac{\hbar \omega}{2} \right)^3 \frac{1}{\text{sh}^2(\frac{\hbar \omega}{2k_B T})} \frac{k_B}{T^2 k_B^2} = \underline{3N k_B \frac{x^2}{\text{sh}^2(x)}}$$

$$\frac{\hbar \omega}{2k_B T} = x \rightarrow \frac{\hbar \omega_e}{2k_B} = T_e$$

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• ravninski rotator ^{ni degenerirano}
↑ _{2x degenerirano}

$$E(j) = k_B T_r j^2; j = 0, 1, 2, \dots$$

$$T_r = 150 \text{ K}$$

a) $T' = 25 \text{ K} \rightarrow T = 30 \text{ K}$

b) $T' = 1500 \text{ K} \rightarrow T = 1505 \text{ K}$

$\Delta S = ?$

$$S(T) = \frac{\langle E \rangle - F}{T}$$

$$e^{-\beta F} = \sum_{j=0}^{\infty} e^{-\beta E(j)} = \sum_{j=0}^{\infty} e^{-\beta k_B T_r j^2} g(j)$$

$= \begin{cases} 1; j=0 \\ 2; \text{ sicer} \end{cases}$

ne znamo rešit analitično

→ visoka T limita, nizka T limita

$$e^{-\beta F} \approx 2 \left(\frac{1}{2} + \int_0^{\infty} e^{-\beta k_B T_r j^2} dj \right) - 1$$
$$= \sqrt{\frac{\pi}{\beta k_B T_r}}$$

$$\Delta S = \frac{k_B}{2} \ln \frac{T}{T'} = 1,4 \cdot 10^{-7} \frac{\text{eV}}{\text{K}}$$

$$e^{-\beta F} \approx 1 + 2e^{-\beta k_B T_r}$$

$$\Delta S = 4 \cdot 10^{-6} \frac{\text{eV}}{\text{K}}$$